

CAMV practice sheet - Answers

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1. A. **True.**

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$$

$$\mathbf{b} \cdot \mathbf{a} = b_x a_x + b_y a_y$$

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$$

B. **True.** The dot product is given by $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$. If the vectors are orthogonal, then $\theta = 90^\circ$ and $\cos 90^\circ = 0$. Thus, $\mathbf{a} \cdot \mathbf{b} = 0$.

C. **False.** They are orthogonal.

2. The y axis is the set of points where $x = 0$. If we want to know where a function intersects the y axis, we need to evaluate it at $x = 0$.

$$f(x) = 2x^4 + 3x - 3$$

$$f(0) = 2(0)^4 + 3(0) - 3 = -3$$

Thus, the function intersects the y axis at the point (0, -3).

False.

3. To determine if a function goes through a point, we need to evaluate the function at the x coordinate of the point and see if it equals the y coordinate of the point.

$$A = (2, 4)$$

$$f(x) = 5x - 6$$

$$f(2) = 5(2) - 6 = 10 - 6 = 4 = A_y$$

Thus, the function goes through the point A.

True.

4. (a) In GeoGebra, type:

$$f(x) = 3x^6 - 2.9x^5 - 4.9x^4 + 2.7x^3 + 0.9x^2 + 1.1x + 0.5$$

(b) Calculate $f(0)$:

$$f(0) = 0.5 \neq 0$$

(c) In GeoGebra, type: $A = (1.23, f(1.23))$.

(d) We are going to use $f'(x)$ to get the derivative of the function. In GeoGebra, type:

$$T = f'(1.23)(x - 1.23) + f(1.23)$$

(e) In GeoGebra, create a slider named x_0 with interval $[-1.5, 2]$ then change the definition of the line T to be:

$$T = f'(x_0)(x - x_0) + f(x_0)$$

(f) In GeoGebra, type: $g(x) = \text{Function}(f, 0.2, 1.5)$.

5. The slope of a line is given by:

$$a = \frac{B_y - A_y}{B_x - A_x}$$

Given the points $A(2, 4)$ and $B(3, -2)$:

$$a = \frac{-2 - 4}{3 - 2} = \frac{-6}{1} = -6$$

6. We simply need to think of point P as a vector with coordinates $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$ and then add the 2 vectors.

$$P + \mathbf{v} = \begin{pmatrix} 1 \\ 6 \end{pmatrix} + \begin{pmatrix} -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1-2 \\ 6+4 \end{pmatrix} = \begin{pmatrix} -1 \\ 10 \end{pmatrix}$$

7. (a) In GeoGebra, type:

$$\text{ImplicitCurve}(x^3 + y^3 - 5xy^2 - x + 1)$$

Then, select the curve, enter the object properties and in the colour tab change the colour to blue.

- (b) No. A simple proof is that, if we plot the curve we can see some x values that have multiple y values as images meaning the equation is not a function and can't be written in the explicit form.
 (c) P is on the curve if its coordinates satisfy the equation of the curve.

$$\begin{aligned} x_P^3 + y_P^3 - 5 \cdot x_P \cdot y_P^2 - x_P + 1 \\ = 1^3 + 4^3 - 5 \cdot 1 \cdot 4^2 - 1 + 1 \\ = -15 \neq 0 \end{aligned}$$

Thus, P is not on the curve and the correct option is **B**.

8. **False.** Here is a counter example:

If $f(x) = x^2 - x$, then $f'(x) = 2x - 1$.

The roots (zeros) of $f(x)$ are $x = 0$ and $x = 1$.

However,

$$\begin{aligned} f'(0) &= -1 \neq 0 \\ f'(1) &= 1 \neq 0 \end{aligned}$$

The real root of $f'(x)$ is $x = \frac{1}{2}$.

Overall, the roots of $f(x)$ don't tell us anything about $f'(x)$.

9. **False.** It can be.

Given 2 points P and Q and $t \in [0, 1]$:

$$\begin{aligned} x(t) &= (1-t)P_x + t \cdot Q_x \\ y(t) &= (1-t)P_y + t \cdot Q_y \end{aligned}$$

or in vector form:

$$\mathbf{r}(t) = (1-t)\mathbf{P} + t \cdot \mathbf{Q}$$

10. A. **True.** Every element of the relation's domain $\{2, 3, 4, 6\}$ is associated to exactly one value.
 B. **False.**
 C. **False.** Elements 4 and 6 of the domain map to the same element 5 of the range.
 D. **True.**
 E. **False.** The domain of the relation F is

$$D_F = \{2, 3, 4, 6\} \neq A = \{1, 2, 3, 4, 5, 6\}$$

- F. **False.** The domain of the relation F is

$$D_F = \{2, 3, 4, 6\} \neq B = \{4, 5, 6, 7, 8\}$$

11. (a) In GeoGebra, type:

$$\begin{aligned} a &= 8.5 \\ b &= 3.9 \end{aligned}$$

Let's define the x and y components of the curve separately:

$$\begin{aligned} cx(t) &= (a-b)\cos(t) + b \cdot \cos((a/b-1)t) \\ cy(t) &= (a-b)\sin(t) - b \cdot \sin((a/b-1)t) \end{aligned}$$

Finally, define the curve:

$$c = \text{Curve}(cx(t), cy(t), t, 0, 12\pi)$$

(b) In GeoGebra, type:

$$t_0 = 9\pi$$

$$P = c(t_0)$$

(c) To calculate the tangent vector, we need to calculate the derivative of each component of the curve at t_0 . Those will be the x and y components of the tangent vector. In GeoGebra, type:

$$\mathbf{u} = (cx'(t_0), cy'(t_0))$$

Now, we need to translate the vector u so that it starts at point P . In GeoGebra, type:

$$\mathbf{v} = \text{Translate}(u, P)$$

Finally, set the vector style to be dashed and hide all the unnecessary objects.

12. (a) In GeoGebra, type: $c = \text{Curve}(\cos(t), \sin(t), t, 0, 2\pi)$

(b) In GeoGebra, type: $P = c(\pi/4)$

(c) In GeoGebra, you can use the graphical tool to put a slider. Set the name to be t_0 and the interval to $[0, \pi]$. Change the definition of point P to be: $P = c(t_0)$.

(d) In GeoGebra, type:

$$c_1 = \text{Curve}(0.5\cos(t) + x(P), 0.5\sin(t) + y(P), t, 0, 2\pi)$$

13. (a) Let's start by defining the points we are interpolating on.

In GeoGebra, type:

$$P_0 = (140, 80)$$

$$P_1 = (60, 80)$$

$$P_2 = (90, 50)$$

$$P_3 = (120, 100)$$

Then, let's define the values t will have at those points

$$t_0 = 0$$

$$t_1 = 1$$

$$t_2 = 2$$

$$t_3 = 3$$

then, let's define the x and y components of the curve.

$$cx = \text{FitPoly}((t_0, x(P_0)), (t_1, x(P_1)), (t_2, x(P_2)), (t_3, x(P_3)), 3)$$

$$cy = \text{FitPoly}((t_0, y(P_0)), (t_1, y(P_1)), (t_2, y(P_2)), (t_3, y(P_3)), 3)$$

Finally, let's define the parametric curve

$$c = \text{Curve}(cx(t), cy(t), t, t_0, t_3)$$

And hide cx and cy .

(b) To determine the tangent vector, we just need to calculate the derivatives of the x and y components when t is at P_3 , that means at t_3 .

In GeoGebra, type:

$$\mathbf{u} = (cx'(t_3), cy'(t_3))$$

Then, shift the vector to start at P_3

$$\mathbf{v} = \text{Translate}(u, P_3)$$

And we finish by hiding the vector $\mathbf{u}$ and changing the color of $\mathbf{v}$ to red and its style to dashed.